

Flanger

Three flangers in one: bucket-brigade pedal, wide-sweep rack and true through-zero.



Overview

Flanger mixes your signal with a copy of itself delayed by a few milliseconds, and sweeps that delay. The two combine into a comb of peaks and notches that moves up and down the spectrum - the swooshing, jet-plane sound.

Three characters are built in. PEDAL is the classic bucket-brigade stompbox: a darker loop with a little grit, made for strong regeneration. RACK is the clean, wide-sweep studio unit, going from metallic flange down to slow, chorus-like movement. THRU-0 recreates the original tape trick, where the swept copy passes right through the dry signal for a deep cancellation at the crossing point.

Besides the usual LFO, the sweep can be frozen so you can place the comb by hand (MATRIX), or driven by how hard you play (ENVELOPE).

Reach for it for

- Jet-plane sweeps on guitars, synths and drum loops
- Through-zero tape flanging on a whole mix or a bus
- Hollow, nasal tones with negative regen
- Static comb-filter colour with MATRIX
- Touch-sensitive swooshes with ENVELOPE
- Wide stereo motion with SPREAD

Controls

Modes (header)

PEDAL	The bucket-brigade pedal. The delay sweeps from about 0.3 to 6 ms, and the loop is darker with the delay chips' faint hiss. High REGEN gives the classic jet sound.
RACK	The wide-sweep studio rack: clean, sweeping from 0.15 to 14 ms. Short settings are metallic; long ones become a slow, chorus-like sweep.
THRU-O	Through-zero flanging, as done with two tape machines. The dry signal is delayed too, and the swept copy passes through it, giving the deep cancellation as it crosses. Best with MIX at 50%.

Flanger

SINE / TRI / SQR / RND	The LFO shape. Triangle is the classic even flanger sweep; sine lingers at the ends; square jumps between two settings; random wanders.
SWEEP	Click to choose what moves the sweep. LFO: the usual automatic sweep. MATRIX: the sweep stops and MANUAL places the comb by hand (the 'filter matrix' trick). ENVELOPE: your playing drives the sweep - louder notes push it further.
FREE / SYNC / FLOW	How the LFO rate is set: FREE in Hz, SYNC locked to the project tempo and timeline, FLOW tempo-synced but running continuously through loops. Not used in MATRIX or ENVELOPE.
RATE / SENS	The LFO speed (0.02 - 10 Hz, or a note value when synced). In ENVELOPE mode this knob becomes SENS: how far your playing pushes the sweep.
MANUAL	Where the sweep sits - the delay time at its centre. Low settings give high, metallic combs; high settings lower, more chorus-like tones.

WIDTH	How far the sweep travels either side of MANUAL. Zero gives a static comb (not used in MATRIX).
REGEN	Feedback, from -95% to +95%. Positive gives the ringing, metallic jet; negative gives a hollow, nasal tone. The level is compensated so strong settings don't jump out.
SPREAD	Offsets the right side's sweep from the left, from 0 to 180 degrees. 0 is a classic mono flange; 90 - 180 gives wide stereo motion.
Display	The live comb filter (frequency response) and the current delay time; MATRIX or ENVELOPE is shown when active.

Drift

A slow random wander, like a hand on the tape reel or an old clock.

RATE	How much the LFO rate wanders.
WIDTH	How much the sweep width wanders.
MANUAL	How much the centre of the sweep wanders.
DRIFT SPEED	How fast the drift moves (0.02 - 2 Hz).

Master

MIX	Dry / flanged balance. 50% gives the deepest notches; lower is subtler; 100% is the delayed signal only (a pitch-wobbling vibrato when sweeping).
OUTPUT	Output level, -24 to +12 dB.
OUTPUT button	Folds the input and output stages out below the panel.
OUT meters	Output level, left and right.

The header

PRESETS	The factory presets. They set the flanger controls; drift speed, output and the input / output stages are left alone.
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	The last entry resets everything.
Theme picker	Changes the panel's colour theme.
S / M / L	Panel size: small, medium or large.
HW	Switches between the hardware-style panel and the flat look.

Input and output stages (OUTPUT fold-out)

TRIM	Input level before anything else, from -24 to +12 dB.
GUARD	On by default. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources. The tag shows how much it is working.
SAT	Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
BIAS	Shapes the saturation: more bias adds more even (2nd) harmonics, for a warmer, rounder colour.
LO BAND / HI BAND	Where the low and high saturation bands start (40 - 400 Hz and 2 - 16 kHz).
CEILING / THRESH	With CLIP: the level the peaks are softly clipped at. With LIMIT: the threshold of a brickwall limiter that holds the peaks at -0.3 dBFS - pulling it down limits harder and makes the output louder.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper (no latency), or a lookahead limiter (3 ms latency, compensated by REAPER).

The ? key

?	Next to the S / M / L size keys. Opens a pop-up with the design credit, the GFX website address and the address of
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this manual. Click anywhere to close it.

Presets

Classic Pedal	The everyday bucket-brigade flanger: triangle sweep, moderate regen.
Jet Plane	Slow, wide sweep with strong regen - the jet.
Negative Hollow	Negative regen for a hollow, nasal sweep.
Stereo Pedal	The pedal with the right side offset by 90 degrees.
Rack Wide Sweep	The rack's long sweep, slow and wide in stereo.
Rack Chorus-Flange	Long delays, gentle width and faster rate: between chorus and flanger.
Thru-Zero Tape	Slow through-zero sweep without regen, like two tape machines.
Thru-Zero Jet	Through-zero with regen for a more intense crossing.
Filter Matrix	The sweep stopped: set the comb by hand with MANUAL.
Envelope Swoosh	Your playing drives the sweep; strong regen.
Synced 1/2 Bar	Rack flanger locked to half-bar sweeps of the project tempo.
Reset all to defaults	Every control back to its starting value.

Tips

- For the classic through-zero effect on a mix bus, use THRU-0 with REGEN at 0, a slow sine sweep and MIX at 50%.
- In MATRIX mode, automate MANUAL for sweeps you control exactly, in time with the music.

- ENVELOPE works well on drums and funky guitar: each hit swooshes, then the comb settles back.
- If the flange sounds too thin or harsh, try negative REGEN or the RACK mode at a higher MANUAL setting.

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